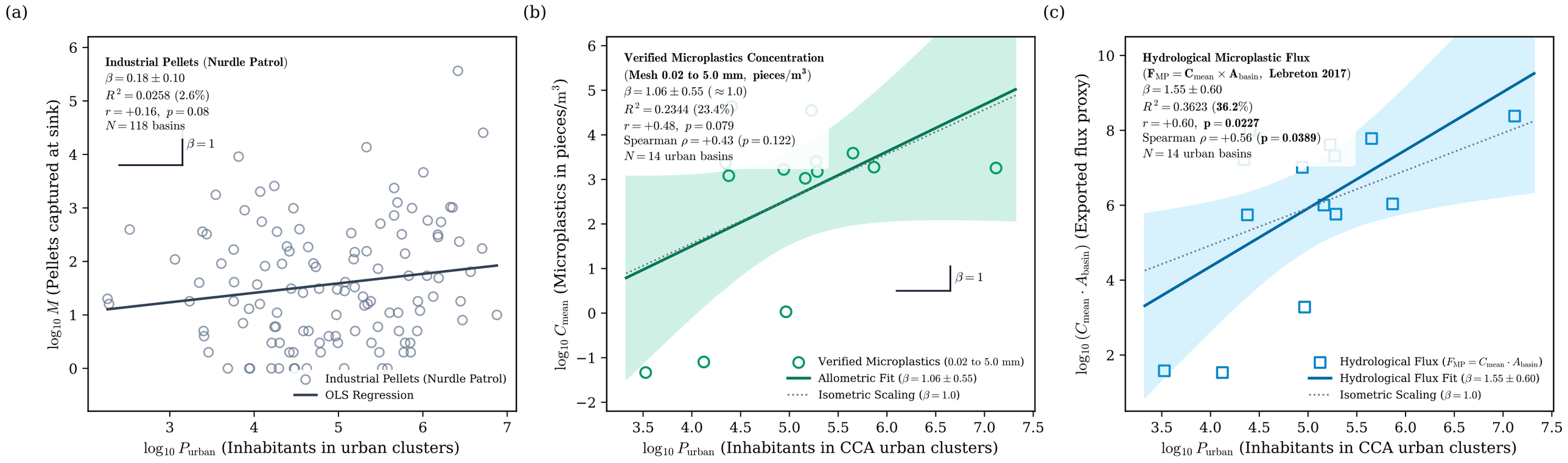

(a)
Industrial Pellets (Nurdle Patrol)
$\beta = 0.18 \pm 0.10$
$R^2 = 0.0258$ (2.6%)
$r = +0.16$, $p = 0.08$
$N = 118$ basins
$\beta = 1$
Industrial Pellets (Nurdle Patrol)
OLS Regression
$\log_{10} M$ (Pellets captured at sink)
$\log_{10} P_{\text{urban}}$ (Inhabitants in urban clusters)
(b)
Verified Microplastics Concentration
(Mesh 0.02 to 5.0 mm, pieces/m^3)
$\beta = 1.06 \pm 0.55$ (≈ 1.0)
$R^2 = 0.2344$ (23.4%)
$r = +0.48$, $p = 0.079$
Spearman $\rho = +0.43$ ($p = 0.122$)
$N = 14$ urban basins
$\beta = 1$
Verified Microplastics (0.02 to 5.0 mm)
Allometric Fit ($\beta = 1.06 \pm 0.55$)
Isometric Scaling ($\beta = 1.0$)
$\log_{10} C_{\text{mean}}$ (Microplastics in pieces/m^3)
$\log_{10} P_{\text{urban}}$ (Inhabitants in CCA urban clusters)
(c)
Hydrological Microplastic Flux
($\mathbf{F_{MP} = C_{mean} \times A_{basin}}$, Lebreton 2017)
$\beta = 1.55 \pm 0.60$
$R^2 = 0.3623$ (**36.2%**)
$r = +0.60$, $\mathbf{p = 0.0227}$
Spearman $\rho = +0.56$ ($\mathbf{p = 0.0389}$)
$N = 14$ urban basins
Hydrological Flux ($F_{MP} = C_{mean} \cdot A_{basin}$)
Hydrological Flux Fit ($\beta = 1.55 \pm 0.60$)
Isometric Scaling ($\beta = 1.0$)
$\log_{10} (C_{\text{mean}} \cdot A_{\text{basin}})$ (Exported flux proxy)
$\log_{10} P_{\text{urban}}$ (Inhabitants in CCA urban clusters)